

Part A. Data

Source: National Oceanic and Atmospheric Administration, Global Monitoring Laboratory

Full URL: https://gml.noaa.gov/webdata/ccgg/trends/co2/co2_annmean_mlo.txt

Description: The dataset contains the annual mean atmospheric carbon dioxide concentration measured at Mauna Loa. I used the observations from 2005 to 2024, giving a total of 20 paired observations. Carbon dioxide concentration is expressed in parts per million.

Variables:

x: number of years since 2005, measured in years

y: annual mean atmospheric CO₂ concentration, measured in ppm

Year	x , years since 2005	CO ₂ , y (ppm)
2005	0	379.98
2006	1	382.09
2007	2	384.02
2008	3	385.83
2009	4	387.64
2010	5	390.10
2011	6	391.85
2012	7	394.06
2013	8	396.74
2014	9	398.81
2015	10	401.01
2016	11	404.41
2017	12	406.76
2018	13	408.72
2019	14	411.65
2020	15	414.21
2021	16	416.41
2022	17	418.53
2023	18	421.08
2024	19	424.61

Part B. Python Script

```
import numpy as np
```

```
import matplotlib.pyplot as plt
```

```
# Annual780, >}</analysis code parameter values from NOAA
```

```
years = np.array([
    2005, 2006, 2007, 2008, 2009,
    2010, 2011, 2012, 2013, 2014,
    2015, 2016, 2017, 2018, 2019,
    2020, 2021, 2022, 2023, 2024
], dtype=float)
```

```
co2 = np.array([
    379.98, 382.09, 384.02, 385.83, 387.64,
    390.10, 391.85, 394.06, 396.74, 398.81,
    401.01, 404.41, 406.76, 408.72, 411.65,
    414.21, 416.41, 418.53, 421.08, 424.61
], dtype=float)
```

```
# x represents the number of years since 2005
```

```
x = years - 2005
```

```
y = co2
```

```
n = len(x)
```

```
print("Number of observations:", n)
```

```
print("x values:", x)
```

```
print("y values:", y)
```

```
# Calculate the summations needed for least squares
```

```
sum_x = np.sum(x)
```

```
sum_y = np.sum(y)
```

```
sum_xy = np.sum(x * y)
```

```
sum_x_squared = np.sum(x ** 2)
```

```
print("\nREQUIRED SUMS")
```

```

print("Sum of x:", sum_x)
print("Sum of y:", sum_y)
print("Sum of xy:", sum_xy)
print("Sum of x squared:", sum_x_squared)

# Calculate the slope using the least-squares formula
a1 = (
    n * sum_xy - sum_x * sum_y
) / (
    n * sum_x_squared - sum_x ** 2
)

print("\nSlope, a1:", a1)

# Calculate the intercept
mean_x = np.mean(x)
mean_y = np.mean(y)

a0 = mean_y - a1 * mean_x

print("Intercept, a0:", a0)
print(f"Regression equation: y = {a0:.4f} + {a1:.4f}x")

# Calculate the fitted CO2 values
y_predicted = a0 + a1 * x

# Residual = measured value - fitted value
residuals = y - y_predicted

print("\nFitted values:")
print(y_predicted)

```

```

print("\nResiduals:")
print(residuals)

# Sum of squared residuals or SSE
Sr = np.sum(residuals ** 2)

# Total sum of squares
St = np.sum((y - mean_y) ** 2)

# Coefficient of determination
r_squared = 1 - (Sr / St)

# Standard error of estimate
standard_error = np.sqrt(Sr / (n - 2))

print("\nREGRESSION STATISTICS")
print(f"SSE, Sr: {Sr:.4f}")
print(f"Total sum of squares, St: {St:.4f}")
print(f"r squared: {r_squared:.4f}")
print(f"Standard error: {standard_error:.4f} ppm")

# Predict the annual mean CO2 for 2025
prediction_year = 2025
prediction_x = prediction_year - 2005

prediction_co2 = a0 + a1 * prediction_x

print("\nPREDICTION")
print(f"x = {prediction_year} - 2005")
print(f"x = {prediction_x}")
print(
    f"y = {a0:.4f} + "

```

```
f"({a1:.4f})({prediction_x})"
)

print(
    f"Predicted CO2 for {prediction_year}: "
    f"{prediction_co2:.2f} ppm")

# Plot the measurements and fitted regression line
plt.figure(figsize=(8, 5))

plt.scatter(
    years,
    y,
    color="royalblue",
    label="Measured CO2"
)

plt.plot(
    years,
    y_predicted,
    color="darkred",
    linewidth=2,
    label="Fitted line"
)

plt.xlabel("Year")
plt.ylabel("Annual Mean CO2 (ppm)")
plt.title("Annual Mean Atmospheric CO2 at Mauna Loa")
plt.grid(True, alpha=0.3)
plt.legend()
plt.tight_layout()

plt.savefig("regression_graph.png", dpi=300)
```

```
plt.show()
```

```
# Plot the residuals
```

```
plt.figure(figsize=(8, 5))
```

```
plt.scatter(  
    years,  
    residuals,  
    color="darkgreen"  
)
```

```
plt.axhline(  
    y=0,  
    color="black",  
    linestyle="--",  
    linewidth=1  
)
```

```
plt.xlabel("Year")  
plt.ylabel("Residual (ppm)")  
plt.title("Residual Plot for the CO2 Regression")  
plt.grid(True, alpha=0.3)  
plt.tight_layout()
```

```
plt.savefig("residual_plot.png", dpi=300)  
plt.show()
```

Part C. Regression Results

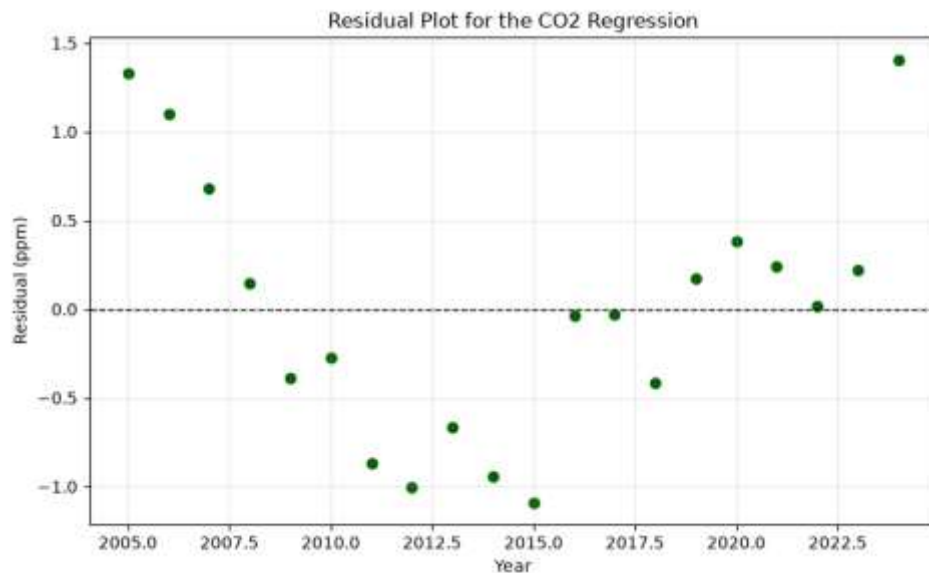
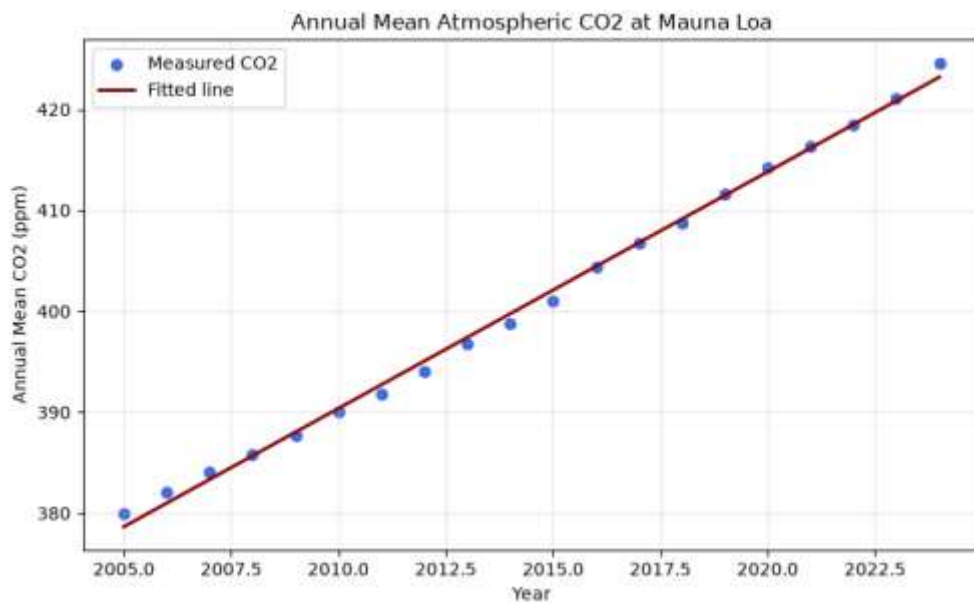
Regression equation

$$y = 378.6469 + 2.3451x$$

where x is the number of years since 2005 and y is the predicted annual mean CO₂ concentration in ppm.

Calculated results

- Intercept: $a_0 = 378.6469$ ppm
- Slope: $a_1 = 2.3451$ ppm/year
- S_r , or SSE: 10.4077 ppm²
- $r^2 = 0.9972$
- Standard error: $s_{y/x} = 0.7604$ ppm



Prediction

For 2025:

$$x = 2025 - 2005 = 20$$

$$y = 378.6469 + 2.3451(20)$$

The predicted annual mean atmospheric CO₂ concentration for 2025 is approximately **425.55 ppm**. This prediction is meaningful because 2025 is only one year beyond the dataset, so it is a short-range extrapolation.

Interpretation

The slope of 2.3451 ppm/year indicates that the annual mean atmospheric CO₂ concentration increased by approximately 2.35 ppm per year during the period covered by the data. The intercept of 378.6469 ppm represents the model's predicted CO₂ concentration when $x = 0$, which corresponds to 2005.

The r^2 value of 0.9972 means that approximately 99.72% of the variation in annual CO₂ concentration is explained by its linear relationship with time. The standard error of 0.7604 ppm indicates that the measured values generally differ from the fitted line by less than one ppm.

The residuals are relatively small and occur on both sides of zero. However, they also show a slight curved pattern, with positive residuals near the beginning and end and mostly negative residuals around the middle. This suggests that the linear model provides a very close approximation for this period, although the actual increase in atmospheric CO₂ is not perfectly constant.